

Derivatives, Rates of Change, and Sensitivity

A field-guide manual for using derivative thinking to measure local change, model response, option Greeks, optimization behavior, portfolio risk, and backtest fragility.

Manual profile

Category: Calculus

Difficulty: Intermediate

Format: Single-column stacked reference-card manual

Audience: Quant research education and meta research-system design

Focus: Derivatives, finite differences, gradients, Greeks, marginal risk, sensitivity diagnostics, and robustness workflows

How to Use This Manual

INTERMEDIATE

READING PATH

Read the first ten pages to lock in derivative mechanics. Then use the middle pages as a sensitivity toolkit for features, portfolios, options, optimization, and backtests.

CORE HABIT

For every page, ask four questions: what is the input, what is the output, what unit is the perturbation in, and does the estimated change remain stable nearby?

INTERMEDIATE FOCUS

This volume assumes you already know the beginner idea of slope. It pushes toward practical market-model use: finite differences, local approximation, partial derivatives, gradients, curvature, and robustness.

IMPLEMENTATION METHOD

Turn formulas into repeatable diagnostic functions. Every sensitivity should record baseline config, perturbed config, sample period, output metric, perturbation size, and rerun hash.

RESEARCH WARNING

Derivative language can sound precise even when the underlying backtest is discontinuous, noisy, or overfit. Treat every slope estimate as a diagnostic, not a law.

BEST USE CASE

Use this manual when reviewing a strategy, debugging a model, interpreting options risk, checking optimization behavior, or building a standardized sensitivity report.

HOW TO USE THIS PAGE

Turn each item into a test: perturb one input, record output change, compare neighboring perturbations, and decide whether the result is robust enough to inform a model decision.

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Notation Reference I

INTERMEDIATE

PURPOSE

This page decodes the symbols used throughout the manual. The important habit is not memorizing notation; it is preserving the input, output, unit, and perturbation meaning behind every symbol.

CORE DERIVATIVE NOTATION

Symbol	Meaning	Watch for
x	input variable	must define units
y or f(x)	model output	metric may be noisy
h	finite step size	too small can amplify noise
Delta x	actual input change	not infinitesimal
dx	ideal differential	model abstraction
f_prime(x)	first derivative	local slope only
f_double_prime(x)	second derivative	curvature, noisy to estimate

UNIT RULE

A derivative inherits units: output units divided by input units. PnL per bp, volatility per day, score per parameter point, and dollars per share are not interchangeable.

AUDIT RULE

Every sensitivity result should state the baseline value, perturbed value, perturbation size, output metric, sample period, and whether the change is causal or diagnostic-only.

HOW TO USE THIS PAGE

Turn each item into a test: perturb one input, record output change, compare neighboring perturbations, and decide whether the result is robust enough to inform a model decision.

Notation Reference II

INTERMEDIATE

MODEL AND PORTFOLIO NOTATION

Symbol	Meaning	Watch for
partial f/partial x	partial derivative	holds other inputs fixed
grad f	gradient vector	direction of steepest local increase
J(theta)	objective/loss	metric design matters
theta	model parameters	overfit risk
Sigma	covariance matrix	regime instability

GREEK-STYLE NOTATION

Options and fixed-income desks often use named sensitivities: delta, gamma, vega, theta, rho, DV01, duration, and convexity. They are domain-specific derivatives or finite-difference approximations.

DISCRETE-MARKET WARNING

Market data is sampled. A one-bar difference is not a true instantaneous derivative. It is a finite-difference estimate whose meaning depends on bar size, data cleaning, and execution timing.

SIGN CONVENTION WARNING

Some sensitivities change sign depending on quoting convention, position direction, or whether the metric is price, loss, return, yield, or cost. State the convention before interpreting the number.

MEMORY ANCHOR

A derivative is not just a formula. It is a controlled question: when this input moves by this amount, what does that output do right here?

HOW TO USE THIS PAGE

Turn each item into a test: perturb one input, record output change, compare neighboring perturbations, and decide whether the result is robust enough to inform a model decision.

Derivative Thinking in the Quant Research Pipeline

INTERMEDIATE

PIPELINE MAP

Raw data -> cleaned prices -> returns -> features -> signals -> positions -> executions -> PnL -> risk metrics -> validation -> deployment.

LOCAL SENSITIVITY LAYER

At each stage, ask what happens when one input is perturbed: a price adjustment, a missing bar, a lookback change, a volatility shock, a fee increase, or a fill delay.

PROPAGATION LAYER

A small change can travel through the pipeline. Price changes affect returns; returns affect features; features affect signals; signals affect positions; positions affect PnL and risk.

DERIVATIVE TOOLS

Use finite differences for black-box backtests, gradients for differentiable models, Greeks for options, marginal contributions for portfolios, and stress ladders for assumptions.

DEPLOYMENT RULE

A strategy is not ready because the baseline score is good. It is closer to ready when nearby perturbations, cost shocks, path changes, and regime splits do not destroy the case.

SYSTEM ARTIFACT

The output of this manual should be a sensitivity report card for every serious model candidate.

HOW TO USE THIS PAGE

Turn each item into a test: perturb one input, record output change, compare neighboring perturbations, and decide whether the result is robust enough to inform a model decision.

Derivative as Local Rate of Change

INTERMEDIATE

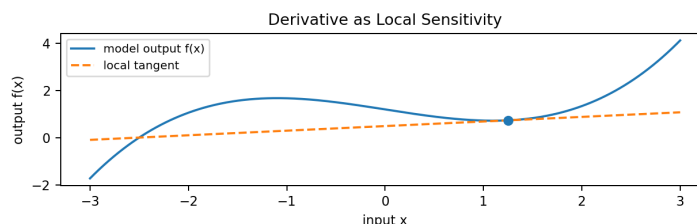
ONE-SENTENCE MEANING

A derivative measures how fast an output changes when one input changes by a tiny amount.

MEMORY JOGGER

Nudge the input; watch the output move.

ANNOTATED CHART



Derivative is the local tangent slope: useful for response around the current point, not a global guarantee.

CORE IDEA

At intermediate level, stop thinking of the derivative as only a school slope formula. Treat it as a local response operator. It tells you the first-order behavior of a model near the current point, not across every regime or every market condition.

QUANT USE + MARKET EXAMPLE

Quant use: Use derivatives to describe signal response, risk exposure, option Greeks, optimization gradients, and marginal portfolio effects.

Example: If model score changes sharply when a lookback changes from 20 to 21 bars, the score has high parameter sensitivity near that setting.

FORMULA / SENSITIVITY RULE

$f_{\text{prime}}(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h$. In sampled markets, you usually estimate this with finite h , not an actual limit.

IMPLEMENTATION NOTE

Create a perturbation function: `run_model(x+h) - run_model(x)`. Store h , baseline output, perturbed output, and the ratio.

SYSTEM FIT + FAILURE MODE

System fit: This is the base primitive for sensitivity diagnostics across features, strategies, risk models, and execution assumptions.

Confusion to avoid: A derivative is local. It does not prove the same response holds far away from the current point.

Related: slope, limit, finite difference, sensitivity, gradient

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Secant Slope vs Tangent Slope

INTERMEDIATE

ONE-SENTENCE MEANING

A secant slope measures average change across an interval; a tangent slope measures local change at a point.

MEMORY JOGGER

Secant is between two points; tangent is at one point.

CORE IDEA

Most market data gives you secants because prices are sampled. A true tangent is a model concept. When you compute one-bar change or five-bar change, you are estimating an average change over a window, not observing an infinitesimal derivative.

QUANT USE + MARKET EXAMPLE

Quant use: Use this distinction when converting price changes, return slopes, factor sensitivities, or backtest metric changes into derivative language. Example: A 20-day trend slope is a secant estimate. It may miss a sudden change that happened on day 19.

FORMULA / SENSITIVITY RULE

Secant: $[f(b)-f(a)]/(b-a)$. Tangent: limit of that ratio as b approaches a .

IMPLEMENTATION NOTE

Compute slopes over multiple window sizes and compare. If the sign flips across windows, the local derivative estimate is unstable.

SYSTEM FIT

This distinction belongs in feature engineering and diagnostics for trend, momentum, and regime detection.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Do not call every difference a derivative without naming the interval length.
Related: finite difference, rolling slope, tangent, local approximation

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Delta Notation and Small Moves

INTERMEDIATE

ONE-SENTENCE MEANING

Delta notation represents a change, while differential notation represents idealized tiny change.

MEMORY JOGGER

Delta is the actual step; d is the calculus limit version.

CORE IDEA

In quant workflows, Delta P , Delta r , Delta w , and Delta θ are real finite perturbations. dP , dr , dw , and $d\theta$ are modeling abstractions. Keeping them separate prevents false precision.

QUANT USE + MARKET EXAMPLE

Quant use: Use deltas for actual backtest perturbation tests; use differentials for model formulas and intuition.

Example: Changing a transaction-cost assumption from 5 bps to 6 bps is Delta $c = 1$ bp, not an infinitesimal dc .

FORMULA / SENSITIVITY RULE

Approximation: $\Delta y \sim f'(x) \Delta x$. Exact equality is not guaranteed unless the function is linear over the move.

IMPLEMENTATION NOTE

Log every perturbation with units: basis points, dollars, percent, bars, shares, or volatility points.

SYSTEM FIT

Delta notation is the audit trail for sensitivity testing and scenario analysis.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Small does not mean harmless. Tiny perturbations can expose unstable thresholds and brittle strategy logic.

Related: differential, perturbation, scenario, local linearity

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Units in Derivatives

INTERMEDIATE

ONE-SENTENCE MEANING

A derivative has output units divided by input units.

MEMORY JOGGER

Slope always carries units.

CORE IDEA

Derivative units are not decorative. They determine whether the sensitivity has operational meaning. Dollars per dollar, PnL per basis point, score per lookback day, volatility per return shock, and risk per weight all mean different things.

QUANT USE + MARKET EXAMPLE

Quant use: Use units to compare sensitivities and avoid mixing incompatible variables.

Example: A strategy losing 0.15 Sharpe per 1 bp of added cost is more interpretable than saying its derivative is -0.15.

FORMULA / SENSITIVITY RULE

Units of $dy/dx = \text{units}(y) / \text{units}(x)$. If y is dollars and x is bps, the derivative is dollars per bp.

IMPLEMENTATION NOTE

Store metadata with every sensitivity result: input name, input unit, perturbation size, output metric, output unit, timestamp, and sample period.

SYSTEM FIT

Unit discipline makes sensitivity reports readable and prevents false comparisons across models.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A dimensionless-looking number can hide a unit error, especially after annualization or percent conversion.

Related: basis points, scaling, annualization, normalization

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Derivative of a Price Curve

INTERMEDIATE

ONE-SENTENCE MEANING

The derivative of price with respect to time is the instantaneous price-change rate in the model.

MEMORY JOGGER

Price slope is dollars per unit time.

CORE IDEA

Price curves in charts look continuous, but observed prices are discrete and noisy. A derivative of price is usually a model-based or smoothed estimate, not a directly observed market object.

QUANT USE + MARKET EXAMPLE

Quant use: Use price slope carefully for trend estimation, but prefer returns for cross-asset comparability.

Example: A stock moving from 100 to 102 over two days has average slope 1 dollar per day, but that says little about percentage growth.

FORMULA / SENSITIVITY RULE

If $P(t)$ is price, dP/dt is price change rate. Return rate often uses $(1/P)(dP/dt)$.

IMPLEMENTATION NOTE

Estimate price slope with rolling regression or finite differences, then test whether signals survive after transaction costs and alternative windows.

SYSTEM FIT

Price derivatives sit in feature engineering, not final performance judgment by themselves.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A high-dollar slope on a high-priced asset may be less meaningful than a lower-dollar slope on a cheaper asset.

Related: return, log return, trend slope, normalization

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Rate of Change vs Return

INTERMEDIATE

ONE-SENTENCE MEANING

Rate of change is raw movement; return scales movement by capital or starting value.

MEMORY JOGGER

Change tells distance; return tells growth rate.

CORE IDEA

A derivative of price can measure raw dollars per time. A return measures proportional growth. For market models, proportional changes are usually more comparable across assets and time periods.

QUANT USE + MARKET EXAMPLE

Quant use: Use returns for portfolio aggregation, risk estimation, and cross-asset comparison; use raw changes when dollars or ticks are the operational unit.

Example: A 2 dollar move means different things for a 20 dollar stock and a 500 dollar stock.

FORMULA / SENSITIVITY RULE

Simple return: $r_t = (P_t - P_{t-1}) / P_{t-1}$. Raw change: $\Delta P_t = P_t - P_{t-1}$.

IMPLEMENTATION NOTE

Implement both but label them clearly. Never feed a model raw price deltas when it expects returns.

SYSTEM FIT

This is a basic feature-definition gate before derivative-style diagnostics.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Do not annualize raw price slopes as if they were returns.

Related: percent change, log return, normalization, scaling

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Log Return as Instantaneous Growth Approximation

INTERMEDIATE

ONE-SENTENCE MEANING

A log return approximates a continuously compounded growth rate over a short interval.

MEMORY JOGGER

Log return adds cleanly across time.

CORE IDEA

When returns are small, $\log(1+r)$ is close to r . In calculus terms, log converts multiplicative growth into additive accumulation and connects discrete observations to continuous growth-rate thinking.

QUANT USE + MARKET EXAMPLE

Quant use: Use log returns for additive time aggregation, volatility modeling, and continuous-time model intuition.

Example: Two daily log returns can be summed to get the two-day log return, which simplifies compounding analysis.

FORMULA / SENSITIVITY RULE

$g_t = \log(P_t/P_{t-1}) = \log(1+r_t)$. For small r , $g_t \sim r_t$.

IMPLEMENTATION NOTE

Compute `np.log(price).diff()`, but handle zero prices, bad ticks, corporate actions, and missing observations.

SYSTEM FIT

Log returns are the bridge from discrete price data to continuous-growth calculus models.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Log return and simple return diverge during large moves; do not assume they are always interchangeable.

Related: continuous compounding, series expansion, additive returns

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

First Differences in Discrete Data

INTERMEDIATE

ONE-SENTENCE MEANING

A first difference is the observed change between adjacent samples.

MEMORY JOGGER

Today minus yesterday.

CORE IDEA

First differences are the practical cousin of first derivatives. They are easy to compute but depend completely on sampling frequency, data cleaning, and alignment.

QUANT USE + MARKET EXAMPLE

Quant use: Use first differences for raw momentum, spread changes, inventory changes, and feature deltas.

Example: $\Delta \text{spread}_t = \text{spread}_t - \text{spread}_{t-1}$ can detect liquidity deterioration.

FORMULA / SENSITIVITY RULE

$\Delta x_t = x_t - x_{t-1}$. Approx derivative = $\Delta x_t / \Delta t$.

IMPLEMENTATION NOTE

In pandas: `series.diff()/dt`. Confirm that `dt` is constant before interpreting the result as a rate.

SYSTEM FIT

First differences are baseline features and diagnostic transforms in many market pipelines.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Differencing a noisy series can make it noisier and less predictive.

Related: finite difference, sampling frequency, microstructure noise

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Forward, Backward, and Central Difference

INTERMEDIATE

ONE-SENTENCE MEANING

Finite differences estimate slope using nearby sampled points.

MEMORY JOGGER

Forward looks ahead, backward looks behind, central uses both sides.

CORE IDEA

Forward differences are natural for simulations but can leak future data in historical features. Backward differences are causal. Central differences are often more accurate for analysis but generally not usable as live trading features.

QUANT USE + MARKET EXAMPLE

Quant use: Choose the difference formula according to whether you are doing offline diagnostics, live signal construction, or model calibration.
Example: A central difference around today uses tomorrow, so it is invalid as a real-time signal.

FORMULA / SENSITIVITY RULE

Forward: $[f(x+h)-f(x)]/h$. Backward: $[f(x)-f(x-h)]/h$. Central: $[f(x+h)-f(x-h)]/(2h)$.

IMPLEMENTATION NOTE

Separate research diagnostics from deployable features. Put leakage checks around any centered calculation.

SYSTEM FIT

Difference choice belongs in feature-definition and backtest-leakage controls.

CONFUSIONS + RELATED TERMS

Confusion to avoid: The most mathematically accurate estimate may be operationally invalid if it uses future data.
Related: look-ahead bias, causal features, numerical derivative

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Step Size h Selection

INTERMEDIATE

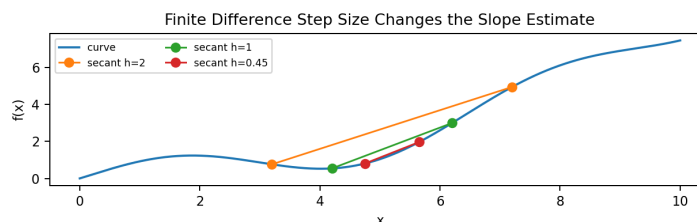
ONE-SENTENCE MEANING

The finite-difference step size controls the tradeoff between approximation error and noise.

MEMORY JOGGER

Too big blurs local behavior; too small magnifies noise.

STEP-SIZE CHART



Changing h changes the finite-difference estimate. Stable slopes across h values are more credible.

CORE IDEA

In exact math, h goes to zero. In numerical work, h is finite. In market data, smaller h may not be better because microstructure noise, discreteness, and threshold effects can dominate.

QUANT USE + MARKET EXAMPLE

Quant use: Use multiple h values to test whether sensitivity estimates are stable.

Example: A strategy score derivative with respect to fee assumptions may change if you test 0.1 bp, 1 bp, and 5 bps perturbations.

FORMULA / SENSITIVITY RULE

Sensitivity _{h} = $[\text{metric}(x+h) - \text{metric}(x)]/h$. Robustness requires similar estimates across a reasonable h range.

IMPLEMENTATION NOTE

Create a small grid of h values and plot sensitivity versus h . Treat wild oscillation as a warning.

SYSTEM FIT + FAILURE MODE

System fit: Step-size sweeps are core infrastructure for robust sensitivity reports.

Confusion to avoid: Tiny perturbations can be meaningless when the system has discrete thresholds, integer shares, minimum ticks, or order-size constraints.

Related: finite difference, numerical stability, perturbation grid

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Noise Amplification by Differentiation

INTERMEDIATE

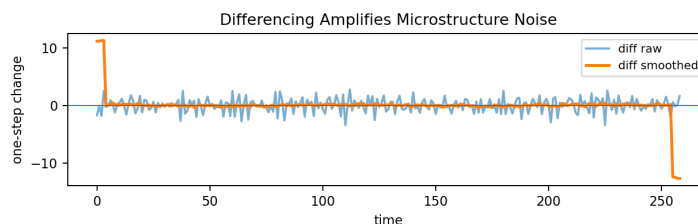
ONE-SENTENCE MEANING

Taking differences often increases the visible impact of random noise.

MEMORY JOGGER

Derivative estimates turn wiggles into spikes.

NOISE CHART



Differencing raw noisy observations can create large apparent changes that are not stable signal.

CORE IDEA

Differentiation is high-pass behavior: it emphasizes rapid changes. That can reveal turning points, but it also magnifies quote noise, bad ticks, and sampling artifacts.

QUANT USE + MARKET EXAMPLE

Quant use: Be careful when creating derivative-like features from high-frequency prices, spreads, or order-book variables.

Example: A stable midprice with bid-ask bounce can produce alternating positive and negative first differences.

FORMULA / SENSITIVITY RULE

If $x_t = \text{signal}_t + \text{noise}_t$, then $\Delta x_t = \Delta \text{signal}_t + \Delta \text{noise}_t$. The noise difference may dominate.

IMPLEMENTATION NOTE

Compare raw differences, smoothed differences, and robust alternatives such as median filters or rolling regression slopes.

SYSTEM FIT + FAILURE MODE

System fit: Noise amplification belongs in data-quality checks and feature robustness testing.

Confusion to avoid: A spiky derivative feature is not automatically informative; it may be mostly market microstructure.

Related: microstructure noise, smoothing, rolling regression, robustness

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Smoothing Before Differentiating

INTERMEDIATE

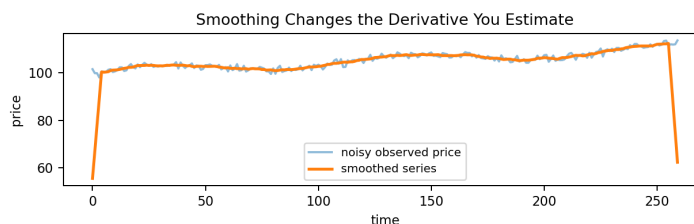
ONE-SENTENCE MEANING

Smoothing can reduce noise before estimating slope, but it also changes timing and shape.

MEMORY JOGGER

Smoothing cleans, but it also delays and distorts.

SMOOTHING CHART



Smoothing reduces noise but shifts timing and can distort local behavior.

CORE IDEA

A smoothed series has different local behavior than the raw series. This may be useful for trend estimation, but it can introduce lag and hide regime breaks.

QUANT USE + MARKET EXAMPLE

Quant use: Use smoothing when the target is slower structure, not when immediate event response or execution timing matters.
Example: A moving-average slope can detect broad trend but may react late to a sharp reversal.

FORMULA / SENSITIVITY RULE

Derivative of smoothed x is not the same as smoothed derivative of x in every pipeline. Order of operations matters.

IMPLEMENTATION NOTE

Backtest raw, smoothed, and alternative smoothing versions side by side. Record lag and turnover impact.

SYSTEM FIT + FAILURE MODE

System fit: Smoothing choice is part of the feature specification and must be versioned.
Confusion to avoid: Do not clean away the exact market behavior your strategy is supposed to exploit.
Related: moving average, EWMA, lag, feature engineering

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Second Derivative as Curvature

INTERMEDIATE

ONE-SENTENCE MEANING

The second derivative measures how the first derivative changes.

MEMORY JOGGER

Slope of the slope.

CORE IDEA

Curvature tells whether the response is accelerating, decelerating, convex, or concave. In finance, curvature is central to options, nonlinear risk, drawdown dynamics, and optimization surfaces.

QUANT USE + MARKET EXAMPLE

Quant use: Use curvature to detect nonlinear payoff behavior, gamma exposure, and unstable objective functions.

Example: A strategy may have stable average return but rapidly worsening losses as volatility rises, showing negative curvature to volatility.

FORMULA / SENSITIVITY RULE

$f_double_prime(x) = d/dx[f_prime(x)]$. Positive curvature bends upward; negative curvature bends downward.

IMPLEMENTATION NOTE

Estimate curvature with second differences, but validate heavily because second differences are very noise-sensitive.

SYSTEM FIT

Curvature diagnostics belong in risk, option exposure, and parameter-surface analysis.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Curvature is not the same as volatility, though both can describe variation in different contexts.

Related: gamma, convexity, Hessian, second difference

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Acceleration of Trend

INTERMEDIATE

ONE-SENTENCE MEANING

Acceleration measures whether the rate of change is itself increasing or decreasing.

MEMORY JOGGER

Is the move speeding up or slowing down?

CORE IDEA

Trend acceleration is a second-order idea. A positive return slope may still be weakening if its rate is falling. This matters for momentum decay and late-entry risk.

QUANT USE + MARKET EXAMPLE

Quant use: Use acceleration-like features to detect exhaustion, breakout follow-through, or reversal risk.

Example: Price keeps rising, but each new five-day return is smaller than the prior one; the trend may be decelerating.

FORMULA / SENSITIVITY RULE

Approx acceleration = $\Delta(\Delta P) / \Delta t^2$ or Delta returns over time.

IMPLEMENTATION NOTE

Use rolling regression slopes of returns instead of raw second differences when noise is high.

SYSTEM FIT

Acceleration features sit in signal refinement and regime diagnostics, not as standalone proof of edge.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Acceleration estimates are fragile; validate out of sample and across assets.

Related: momentum, second difference, curvature, regime shift

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Convexity in Payoffs

INTERMEDIATE

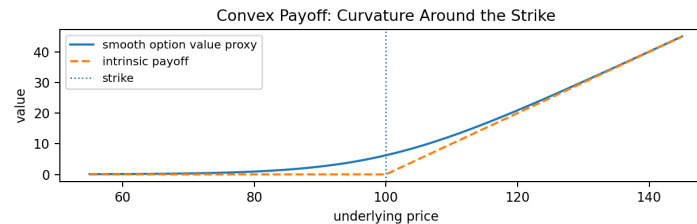
ONE-SENTENCE MEANING

A convex payoff gains sensitivity as the underlying moves in a favorable direction.

MEMORY JOGGER

The curve bends upward.

PAYOFF CHART



Convex payoff curvature means first-order exposure changes as the underlying moves.

CORE IDEA

Convexity means the slope changes in your favor as the input rises, or losses may be limited while gains expand. Options are the classic example, but convexity also appears in stop rules, trend following, and nonlinear risk controls.

QUANT USE + MARKET EXAMPLE

Quant use: Use convexity language to understand nonlinear PnL, option exposure, and asymmetric strategy behavior.

Example: A call option becomes more sensitive to stock price as the stock moves through the strike region.

FORMULA / SENSITIVITY RULE

Convex if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$. Locally, positive second derivative suggests convexity.

IMPLEMENTATION NOTE

Plot PnL against the underlying driver. Estimate slope in different zones instead of relying on one average beta.

SYSTEM FIT + FAILURE MODE

System fit: Convexity analysis belongs in payoff mapping and stress testing.

Confusion to avoid: Convex payoff does not mean guaranteed profit; premium, timing, and path matter.

Related: gamma, nonlinear exposure, payoff diagram, option value

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Concavity and Saturation

INTERMEDIATE

ONE-SENTENCE MEANING

A concave response increases at a decreasing rate or saturates after a point.

MEMORY JOGGER

More input still helps, but each extra unit helps less.

CORE IDEA

Concavity shows diminishing marginal benefit. Many strategy knobs behave this way: more leverage, more smoothing, more diversification, or more training data may help until the incremental gain fades.

QUANT USE + MARKET EXAMPLE

Quant use: Use concavity to detect diminishing returns and avoid over-tuning past the useful region.

Example: Increasing lookback from 5 to 20 days stabilizes a signal, but increasing from 80 to 95 days adds little.

FORMULA / SENSITIVITY RULE

Negative second derivative indicates local concavity: $f_{\text{double_prime}}(x) < 0$.

IMPLEMENTATION NOTE

Plot metric versus parameter and look for flattening. Penalize choices that chase tiny extra gains in flat regions.

SYSTEM FIT

Concavity diagnostics support parameter selection and resource allocation.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A flat response region can indicate robustness or simply an insensitive metric; interpret with domain context.

Related: diminishing returns, saturation, curvature, parameter robustness

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Partial Derivative

INTERMEDIATE

ONE-SENTENCE MEANING

A partial derivative measures sensitivity to one input while holding other inputs fixed.

MEMORY JOGGER

Move one knob; freeze the rest.

CORE IDEA

Most quant functions have multiple inputs. Partial derivatives isolate one local relationship, but this isolation can be artificial when variables are correlated or constrained in reality.

QUANT USE + MARKET EXAMPLE

Quant use: Use partial derivatives for portfolio weights, model parameters, option variables, and risk factors.

Example: Option delta changes stock price while holding volatility and time fixed; vega changes volatility while holding price and time fixed.

FORMULA / SENSITIVITY RULE

Partial $f / \text{partial } x_i$ = local change in f when x_i changes and other inputs are held fixed.

IMPLEMENTATION NOTE

Perturb one column, parameter, or scenario variable at a time. Record which variables were frozen.

SYSTEM FIT

Partial derivatives are the building block of sensitivity matrices and Greeks.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Holding everything else fixed may be unrealistic if variables co-move in real markets.

Related: gradient, Jacobian, Greeks, ceteris paribus

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Gradient Vector

INTERMEDIATE

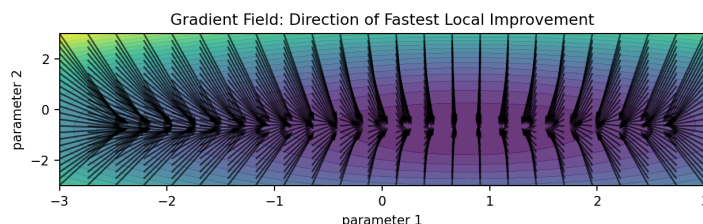
ONE-SENTENCE MEANING

A gradient collects all first partial derivatives into one vector.

MEMORY JOGGER

All the local slopes packed together.

GRADIENT FIELD



Gradient vectors show local direction of fastest improvement or decline.

CORE IDEA

For a multi-parameter objective, the gradient points in the direction of steepest local increase. In minimization, you often move against it.

QUANT USE + MARKET EXAMPLE

Quant use: Use gradients for optimization, calibration, differentiable models, and parameter sensitivity ranking.

Example: A loss function over two parameters can show whether changing lookback or threshold would reduce error faster.

FORMULA / SENSITIVITY RULE

$\text{grad } f(\theta) = [\text{partial } f/\text{partial } \theta_1, \dots, \text{partial } f/\text{partial } \theta_n]$.

IMPLEMENTATION NOTE

For black-box backtests, estimate gradients numerically; for differentiable models, use automatic differentiation.

SYSTEM FIT + FAILURE MODE

System fit: Gradient vectors connect sensitivity testing to optimization engines.

Confusion to avoid: A gradient is local and can point toward a poor solution if the surface is noisy or nonconvex.

Related: partial derivative, optimization, descent direction, objective function

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Directional Derivative

INTERMEDIATE

ONE-SENTENCE MEANING

A directional derivative measures change in a specific direction through input space.

MEMORY JOGGER

Not every move is one knob; sometimes you move along a path.

CORE IDEA

In correlated markets or constrained portfolios, inputs often move together. A directional derivative measures sensitivity along a vector rather than along a single axis.

QUANT USE + MARKET EXAMPLE

Quant use: Use directional derivatives for scenario shocks, factor moves, yield-curve shifts, or portfolio rebalances.
Example: A risk-off scenario may move equities down, volatility up, and credit spreads wider at the same time.

FORMULA / SENSITIVITY RULE

Directional derivative in unit direction u : $D_u f = \text{grad } f \cdot u$.

IMPLEMENTATION NOTE

Define scenario vectors and perturb the whole input vector together. Compare to one-at-a-time sensitivities.

SYSTEM FIT

Directional derivatives belong in stress testing and macro/factor scenario analysis.

CONFUSIONS + RELATED TERMS

Confusion to avoid: One-axis partial derivatives can understate risk when real shocks are coordinated.
Related: gradient, scenario vector, factor shock, stress test

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Jacobian Matrix

INTERMEDIATE

ONE-SENTENCE MEANING

A Jacobian stores first derivatives for a vector-output function.

MEMORY JOGGER

Rows are outputs; columns are input sensitivities.

CORE IDEA

Many research pipelines output multiple values: return forecast, risk estimate, position size, turnover, and confidence. The Jacobian shows how each output responds to each input.

QUANT USE + MARKET EXAMPLE

Quant use: Use Jacobians to diagnose feature pipelines, multi-output models, and risk systems.

Example: A feature shock may increase expected return but also increase turnover and cost exposure.

FORMULA / SENSITIVITY RULE

$J_{ij} = \text{partial output}_i / \text{partial input}_j$. It is a local sensitivity map.

IMPLEMENTATION NOTE

Build a sensitivity matrix by perturbing each input and recording all relevant outputs.

SYSTEM FIT

Jacobians are useful for system-level diagnostics beyond a single score.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A single performance metric can hide bad movement in secondary outputs.

Related: gradient, sensitivity matrix, multi-output model, pipeline

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Hessian Matrix

INTERMEDIATE

ONE-SENTENCE MEANING

A Hessian stores second partial derivatives and cross-curvatures.

MEMORY JOGGER

Curvature map for a multi-input surface.

CORE IDEA

The Hessian describes how gradients change. It can reveal bowls, ridges, saddle points, and parameter interactions. In noisy backtests, Hessian estimates are often unstable but still conceptually useful.

QUANT USE + MARKET EXAMPLE

Quant use: Use Hessian intuition to understand optimizer behavior and parameter-surface fragility.

Example: A strategy can be stable to fast lookback alone and slow lookback alone but unstable when both change together.

FORMULA / SENSITIVITY RULE

$H_{ij} = \partial^2 f / \partial \theta_i \partial \theta_j$. Diagonal terms are own curvature; off-diagonal terms are interaction curvature.

IMPLEMENTATION NOTE

Estimate cautiously with parameter grids. Prefer visual surfaces and stability regions over exact Hessian numbers for backtests.

SYSTEM FIT

Hessian thinking supports optimization diagnostics and robustness analysis.

CONFUSIONS + RELATED TERMS

Confusion to avoid: An estimated Hessian from a noisy objective can be mostly noise.

Related: curvature, cross-partial, saddle point, response surface

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Chain Rule in Research Pipelines

INTERMEDIATE

ONE-SENTENCE MEANING

The chain rule describes sensitivity through nested transformations.

MEMORY JOGGER

Output changes because each stage passes change forward.

CORE IDEA

A quant pipeline is usually nested: raw data -> features -> forecasts -> signals -> positions -> PnL. The chain rule explains how a small input perturbation propagates through every stage.

QUANT USE + MARKET EXAMPLE

Quant use: Use chain-rule thinking to debug why a small data or assumption change creates a large final PnL change.

Example: A small price adjustment changes returns, which changes volatility, which changes position sizing, which changes PnL.

FORMULA / SENSITIVITY RULE

If $y = f(g(x))$, then $dy/dx = f_{\text{prime}}(g(x)) * g_{\text{prime}}(x)$.

IMPLEMENTATION NOTE

Instrument intermediate outputs. Perturb raw inputs and record changes at each stage, not just final backtest score.

SYSTEM FIT

This is the sensitivity backbone of a research system.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Final output sensitivity can come from an intermediate threshold or sizing rule, not the original feature itself.

Related: composition, pipeline, Jacobian, feature dependency

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Product Rule for PnL Components

INTERMEDIATE

ONE-SENTENCE MEANING

The product rule handles change in a product of two changing quantities.

MEMORY JOGGER

When both parts move, count both effects.

CORE IDEA

PnL often looks like position times price change. If both position and price are functions of time or state, total change has contributions from both components.

QUANT USE + MARKET EXAMPLE

Quant use: Use product-rule thinking for dynamic positions, leverage, exposure, and mark-to-market changes.
Example: A strategy can gain because price moved favorably, because position size increased, or both.

FORMULA / SENSITIVITY RULE

$d(uv) = u dv + v du$. For PnL-like terms: change in position value depends on price move and position change.

IMPLEMENTATION NOTE

Decompose PnL into price effect, sizing effect, and interaction/rebalancing effect.

SYSTEM FIT

Product decomposition improves attribution and debugging.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Do not attribute all PnL to signal quality when position sizing changed at the same time.
Related: PnL attribution, exposure, rebalancing, leverage

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Quotient Rule and Ratio Metrics

INTERMEDIATE

ONE-SENTENCE MEANING

The quotient rule handles sensitivity of ratios.

MEMORY JOGGER

Ratios move because numerator, denominator, or both move.

CORE IDEA

Sharpe, information ratio, turnover-adjusted return, hit-rate ratios, and cost ratios all have numerator-denominator behavior. A ratio can improve because the numerator rises or because the denominator falls.

QUANT USE + MARKET EXAMPLE

Quant use: Use quotient-rule thinking when interpreting performance metrics and risk-adjusted scores.

Example: Sharpe can rise because returns improve or because volatility estimate declines. Those are different stories.

FORMULA / SENSITIVITY RULE

$d(u/v) = (v \, du - u \, dv)/v^2$. Denominator sensitivity is especially dangerous when v is small.

IMPLEMENTATION NOTE

Report numerator and denominator changes beside the ratio. Avoid ranking systems by ratio alone.

SYSTEM FIT

Quotient diagnostics belong in model evaluation and performance reporting.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A great ratio with a tiny denominator can be fragile or meaningless.

Related: Sharpe ratio, information ratio, denominator risk, metric design

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Elasticity

INTERMEDIATE

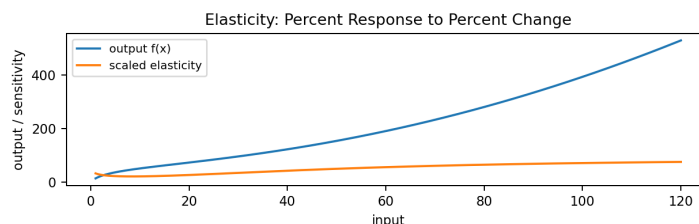
ONE-SENTENCE MEANING

Elasticity measures percent output change per percent input change.

MEMORY JOGGER

Percent response to percent nudge.

ELASTICITY CHART



Elasticity rescales derivative information into percent response.

CORE IDEA

Elasticity is useful when absolute units make sensitivities hard to compare. It rescales derivative information into proportional response.

QUANT USE + MARKET EXAMPLE

Quant use: Use elasticity for demand-like relationships, liquidity response, cost response, and scale effects.

Example: If doubling order size more than doubles slippage, slippage has high elasticity to size.

FORMULA / SENSITIVITY RULE

Elasticity = $(dy/dx) * (x/y)$, when x and y are positive and meaningful as ratios.

IMPLEMENTATION NOTE

Estimate log-log slopes for positive variables: $\Delta \log(y) / \Delta \log(x)$.

SYSTEM FIT + FAILURE MODE

System fit: Elasticity helps compare sensitivity across differently scaled assets or trading sizes.

Confusion to avoid: Elasticity is not safe when y is near zero, negative, or not meaningfully proportional.

Related: log-log model, scaling, market impact, proportional change

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Local Linear Approximation

INTERMEDIATE

ONE-SENTENCE MEANING

A derivative gives a straight-line approximation near the current point.

MEMORY JOGGER

Near here, pretend the curve is a line.

CORE IDEA

The first-order approximation is the practical workhorse of sensitivity. It says the output change is roughly derivative times input change. The approximation degrades as moves get larger or curvature gets stronger.

QUANT USE + MARKET EXAMPLE

Quant use: Use local linear approximations for quick risk estimates and small scenario shocks.

Example: Delta hedging uses a first-order approximation of option value change with respect to underlying price.

FORMULA / SENSITIVITY RULE

$f(x+h) \sim f(x) + f_{\text{prime}}(x)h$. Error depends on curvature and move size.

IMPLEMENTATION NOTE

Validate approximation by comparing predicted change to actual rerun change for several h values.

SYSTEM FIT

Local approximation is fast but must be paired with error checks.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Linear approximation can fail badly across jumps, thresholds, and convex payoffs.

Related: Taylor series, delta, approximation error, curvature

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

First-Order Taylor Expansion

INTERMEDIATE

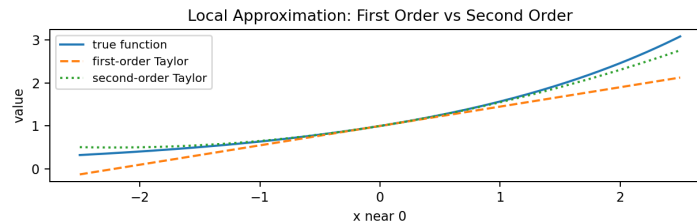
ONE-SENTENCE MEANING

A first-order Taylor expansion approximates a function with its value plus local slope.

MEMORY JOGGER

Base value plus slope times move.

TAYLOR CHART



First-order approximation is fast and local; second order adds curvature.

CORE IDEA

Taylor expansion formalizes local approximation. First order keeps only the slope term, which is often enough for small risk shocks but not enough for highly curved instruments or strategy rules.

QUANT USE + MARKET EXAMPLE

Quant use: Use first-order Taylor logic for delta exposure, factor sensitivity, and small perturbation analysis.

Example: Portfolio value under a small factor shock can be approximated by current value plus factor beta times shock.

FORMULA / SENSITIVITY RULE

$f(x+h) = f(x) + f_{\text{prime}}(x)h + \text{higher-order terms}$. First order drops the rest.

IMPLEMENTATION NOTE

Build sensitivity-based scenario estimates, then compare with full repricing or full backtest reruns.

SYSTEM FIT + FAILURE MODE

System fit: First-order estimates are speed tools for risk and diagnostics.

Confusion to avoid: Dropping higher-order terms is an assumption, not a fact.

Related: Taylor series, delta, linearization, local model

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Second-Order Taylor Expansion

INTERMEDIATE

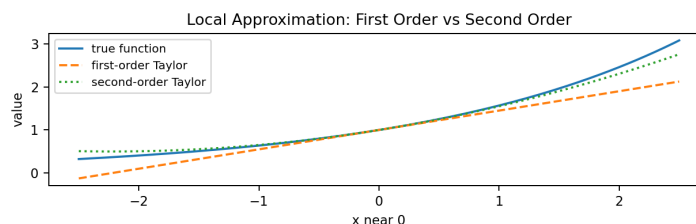
ONE-SENTENCE MEANING

A second-order approximation adds curvature to the local slope estimate.

MEMORY JOGGER

Line plus bend.

TAYLOR CHART



Second-order approximation improves curved local behavior if curvature is estimated reliably.

CORE IDEA

Second order matters when the function bends meaningfully near the current point. Option gamma, bond convexity, and nonlinear cost curves are common finance examples.

QUANT USE + MARKET EXAMPLE

Quant use: Use second-order terms when first-order sensitivity underestimates risk or opportunity.

Example: A delta-only option estimate may miss large value changes when the underlying moves near the strike.

FORMULA / SENSITIVITY RULE

$$f(x+h) \sim f(x) + f_{\text{prime}}(x)h + 0.5 \cdot f_{\text{double_prime}}(x) \cdot h^2.$$

IMPLEMENTATION NOTE

For stress scenarios, compare first-order, second-order, and full-revaluation estimates.

SYSTEM FIT + FAILURE MODE

System fit: Second-order approximation improves risk estimation for nonlinear exposures.

Confusion to avoid: Second-order estimates need reliable curvature; noisy curvature can mislead.

Related: gamma, convexity, Hessian, approximation error

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Approximation Error

INTERMEDIATE

ONE-SENTENCE MEANING

Approximation error is the gap between the derivative-based estimate and the actual function value.

MEMORY JOGGER

The model shortcut has a bill.

CORE IDEA

Derivative approximations are useful because they are fast and interpretable, but the error grows with curvature, discontinuities, large shocks, and unstable systems.

QUANT USE + MARKET EXAMPLE

Quant use: Measure error whenever using local sensitivities for risk, sizing, or model selection.

Example: Predicted PnL change from cost sensitivity may be too small if a fee increase flips trades below a threshold.

FORMULA / SENSITIVITY RULE

Error = actual $f(x+h)$ - approximated $f(x+h)$. Track error across h values.

IMPLEMENTATION NOTE

Create approximation-vs-rerun tests for sensitivities used in dashboards or risk alerts.

SYSTEM FIT

Approximation error is the quality-control metric for derivative-based reports.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A sensitivity number without error testing can create false confidence.

Related: Taylor remainder, validation, threshold effects, stress testing

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Numerical Sensitivity Testing

INTERMEDIATE

ONE-SENTENCE MEANING

Numerical sensitivity testing reruns a model after controlled perturbations.

MEMORY JOGGER

Change one thing; rerun; measure the difference.

CORE IDEA

Many quant systems are black boxes with branching logic, fills, fees, and data transformations. Numerical sensitivity is often more realistic than symbolic differentiation.

QUANT USE + MARKET EXAMPLE

Quant use: Use numerical tests for backtests, execution models, parameter choices, and stress assumptions.

Example: Rerun the same strategy with fees of 3, 5, 7, and 10 bps to see performance slope versus cost.

FORMULA / SENSITIVITY RULE

$\text{Sensitivity} = [\text{metric}(\text{perturbed}) - \text{metric}(\text{base})] / \text{perturbation}.$

IMPLEMENTATION NOTE

Automate perturbation batches and save config hashes so every sensitivity result is reproducible.

SYSTEM FIT

This is the practical derivative engine for non-differentiable research systems.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A numerical derivative can be contaminated by randomness unless seeds, samples, and execution assumptions are controlled.

Related: finite difference, scenario analysis, reproducibility, backtest rerun

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Parameter Perturbation

INTERMEDIATE

ONE-SENTENCE MEANING

Parameter perturbation measures how outputs change when model settings are nudged.

MEMORY JOGGER

Bump the knob and see what breaks.

CORE IDEA

A robust model should not depend on one magical parameter value. Perturbing parameters reveals whether performance is structurally stable or concentrated around a lucky setting.

QUANT USE + MARKET EXAMPLE

Quant use: Use perturbations for lookbacks, thresholds, decay rates, stop distances, and sizing parameters.

Example: A moving-average crossover strategy that only works at 17/43 and fails at 16/43 or 18/43 is suspect.

FORMULA / SENSITIVITY RULE

Parameter sensitivity_i = Delta score / Delta theta_i. Compare across neighboring theta values.

IMPLEMENTATION NOTE

Run local grids around chosen parameters and summarize slope, curvature, and sign stability.

SYSTEM FIT

Parameter perturbation belongs in model validation before deployment.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A single best parameter is less informative than the neighborhood around it.

Related: grid search, robustness, overfitting, response surface

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

One-at-a-Time Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

One-at-a-time sensitivity perturbs one input while holding others fixed.

MEMORY JOGGER

Simple, readable, incomplete.

CORE IDEA

This approach is easy to interpret but can miss interactions. It is useful as a first diagnostic, not as the entire robustness process.

QUANT USE + MARKET EXAMPLE

Quant use: Use it to rank which assumptions or parameters matter most.

Example: Test fee, slippage, borrow cost, delay, and fill-rate assumptions independently before testing combined scenarios.

FORMULA / SENSITIVITY RULE

$S_i = [M(x_i + h_i, \text{others fixed}) - M(\text{base})] / h_i$.

IMPLEMENTATION NOTE

Generate a tornado chart or ranked sensitivity table from one-at-a-time perturbations.

SYSTEM FIT

This is a first-pass sensitivity report for research dashboards.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Real markets rarely move one assumption at a time.

Related: partial derivative, tornado chart, scenario testing, interactions

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Grid Sensitivity

INTERMEDIATE

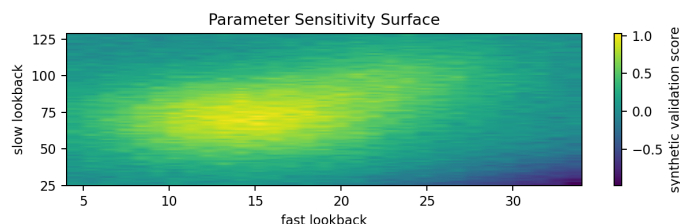
ONE-SENTENCE MEANING

Grid sensitivity tests multiple parameter values over a surface or lattice.

MEMORY JOGGER

Map the neighborhood, not just the point.

PARAMETER HEATMAP



A broad stable region is usually more credible than one isolated high score.

CORE IDEA

A parameter grid reveals ridges, plateaus, cliffs, and isolated peaks. It is more informative than a single local derivative when the system is nonlinear or discontinuous.

QUANT USE + MARKET EXAMPLE

Quant use: Use grids to evaluate lookbacks, thresholds, stops, leverage caps, and model hyperparameters.

Example: A broad high-performing region across nearby fast/slow moving averages is more credible than a lone hot pixel.

FORMULA / SENSITIVITY RULE

Evaluate $\text{score}(\theta_1, \theta_2, \dots)$ over a grid; inspect local slopes and curvature visually.

IMPLEMENTATION NOTE

Plot heatmaps and compute neighborhood statistics: median score, rank stability, and local drawdown dispersion.

SYSTEM FIT + FAILURE MODE

System fit: Grid sensitivity is central to overfit detection.

Confusion to avoid: A dense grid can still overfit if repeatedly mined without honest validation.

Related: response surface, heatmap, parameter robustness, validation

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Response Surface

INTERMEDIATE

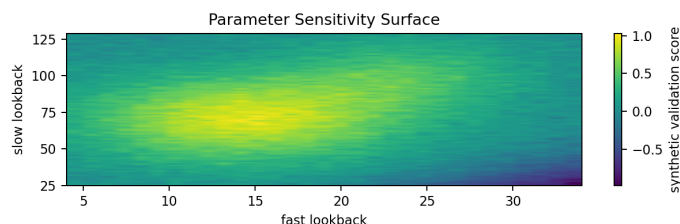
ONE-SENTENCE MEANING

A response surface shows how a metric changes across input or parameter values.

MEMORY JOGGER

Turn model behavior into terrain.

RESPONSE SURFACE HEATMAP



Surface terrain reveals ridges, plateaus, cliffs, and fragile peaks.

CORE IDEA

Instead of reading a single score, view the objective as a surface. Peaks, valleys, flat regions, ridges, and cliffs reveal how sensitive the model is to choices.

QUANT USE + MARKET EXAMPLE

Quant use: Use response surfaces for parameter tuning, risk assumptions, execution costs, and portfolio weights.

Example: A Sharpe surface over entry threshold and exit threshold may show whether the strategy has a stable design zone.

FORMULA / SENSITIVITY RULE

Surface: $z = \text{score}(\theta_1, \theta_2)$. Local derivative is the surface slope near a chosen point.

IMPLEMENTATION NOTE

Create heatmaps with out-of-sample metrics, not just in-sample scores. Use consistent color scales.

SYSTEM FIT + FAILURE MODE

System fit: Response surfaces turn derivative intuition into visual model diagnostics.

Confusion to avoid: Pretty heatmaps can hide low trade count, high turnover, or unstable regimes.

Related: heatmap, gradient, Hessian, robustness

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Stable Region vs Isolated Peak

INTERMEDIATE

ONE-SENTENCE MEANING

A stable region is a broad area of acceptable performance; an isolated peak is a fragile optimum.

MEMORY JOGGER

Prefer a hilltop plateau over a needle.

CORE IDEA

Derivative thinking says steep slopes around an optimum indicate high sensitivity. If a small parameter move destroys performance, the selected parameter may be overfit.

QUANT USE + MARKET EXAMPLE

Quant use: Use stability regions to choose parameters with lower fragility, even if the absolute score is slightly lower.
Example: A Sharpe of 1.4 in a broad region may be better than a Sharpe of 1.7 at one isolated parameter pair.

FORMULA / SENSITIVITY RULE

Fragility can be approximated by local gradient magnitude or score drop to neighboring grid cells.

IMPLEMENTATION NOTE

Rank candidates by neighborhood median and worst-neighbor score, not only maximum score.

SYSTEM FIT

This is a model-selection rule for practical deployment.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Best historical score is not the same as best deployable setting.
Related: overfitting, robustness, parameter surface, local gradient

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Gradient Descent

INTERMEDIATE

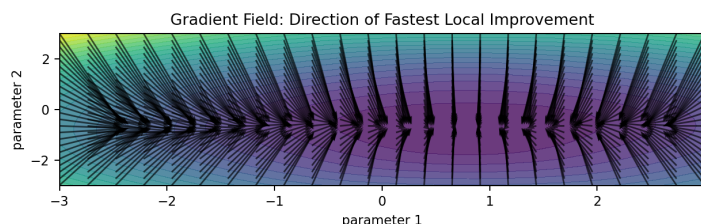
ONE-SENTENCE MEANING

Gradient descent updates parameters by moving opposite the gradient of a loss function.

MEMORY JOGGER

Walk downhill on the loss surface.

GRADIENT FIELD



Descent follows local slope information; noisy surfaces can mislead the path.

CORE IDEA

Gradient descent is the core optimization idea behind many machine learning models. It assumes local slope information can guide better parameters, but noisy surfaces, poor scaling, and bad learning rates can break it.

QUANT USE + MARKET EXAMPLE

Quant use: Use gradient descent for fitting differentiable models, neural nets, and regularized objective functions.

Example: A model adjusts coefficients to reduce forecasting error on returns or volatility targets.

FORMULA / SENSITIVITY RULE

$\theta_{\text{new}} = \theta_{\text{old}} - \eta * \text{grad } J(\theta)$. Eta is the learning rate.

IMPLEMENTATION NOTE

Track loss curves, gradient norms, validation loss, and parameter drift during training.

SYSTEM FIT + FAILURE MODE

System fit: Gradient descent connects calculus to model fitting infrastructure.

Confusion to avoid: Lower training loss does not guarantee better trading performance.

Related: optimization, learning rate, loss function, gradient

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Learning Rate as Step Size

INTERMEDIATE

ONE-SENTENCE MEANING

The learning rate controls how far an optimizer moves along the gradient direction.

MEMORY JOGGER

Step too small, crawl; step too large, bounce.

CORE IDEA

A learning rate is not the derivative itself, but it determines how derivative information gets used. Bad step size can make an otherwise valid gradient useless.

QUANT USE + MARKET EXAMPLE

Quant use: Tune learning rates carefully for predictive models and calibration routines.

Example: A volatility model may fail to converge because the optimizer overshoots the minimum repeatedly.

FORMULA / SENSITIVITY RULE

Update size = η * gradient. Both η and gradient scale matter.

IMPLEMENTATION NOTE

Use learning-rate schedules, gradient clipping, and validation monitoring. Standardize features before optimization.

SYSTEM FIT

Learning-rate diagnostics belong in ML training logs and calibration tools.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Changing learning rate can change optimization behavior without changing the underlying objective.

Related: gradient descent, convergence, optimizer, scaling

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Flat Gradients

INTERMEDIATE

ONE-SENTENCE MEANING

A flat gradient means the output changes very little for local input moves.

MEMORY JOGGER

The surface is nearly level.

CORE IDEA

Flatness can mean robustness, saturation, poor signal, or a model stuck in an uninformative region. Context decides whether flat sensitivity is good or bad.

QUANT USE + MARKET EXAMPLE

Quant use: Use flat-gradient checks when a parameter appears irrelevant or a model stops learning.

Example: A threshold parameter from 0.30 to 0.40 may produce the same trades, so performance barely changes.

FORMULA / SENSITIVITY RULE

If $|\partial f / \partial \theta| \sim 0$, local sensitivity is small.

IMPLEMENTATION NOTE

Check whether the output truly stays stable or whether the metric is too coarse to show changes.

SYSTEM FIT

Flat-gradient interpretation belongs in parameter diagnostics.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Flat does not always mean safe; it may mean the model is insensitive because it is inactive.

Related: saturation, plateau, dead feature, robustness

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Exploding or Unstable Gradients

INTERMEDIATE

ONE-SENTENCE MEANING

An unstable gradient changes wildly across nearby points or steps.

MEMORY JOGGER

The slope estimate is jumping around.

CORE IDEA

Large or oscillating gradients can come from bad scaling, discontinuous rules, noisy objectives, or overly small finite-difference steps.

QUANT USE + MARKET EXAMPLE

Quant use: Watch for unstable gradients in machine learning, backtest optimization, and sensitivity dashboards.

Example: Small changes in stop-loss distance produce huge PnL swings because they alter a few major trades.

FORMULA / SENSITIVITY RULE

Instability symptom: sensitivity_h varies sharply across nearby h or θ .

IMPLEMENTATION NOTE

Standardize inputs, clip gradients where appropriate, increase sample size, or replace cliff-like rules with smoother alternatives.

SYSTEM FIT

Gradient stability is a quality metric for optimization readiness.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A huge derivative is not automatically alpha; it may be fragile dependence.

Related: gradient norm, clipping, discontinuity, objective noise

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Threshold Effects and Discontinuities

INTERMEDIATE

ONE-SENTENCE MEANING

A threshold can create jumps where small input changes cause large output changes.

MEMORY JOGGER

One tiny nudge can flip the switch.

CORE IDEA

Many trading systems use if-statements: enter above threshold, exit below threshold, cap position, stop out, reject illiquid trades. These rules can make derivatives undefined or misleading.

QUANT USE + MARKET EXAMPLE

Quant use: Identify thresholds before applying smooth calculus intuition.

Example: A signal at 0.499 creates no trade while 0.501 creates a full position. Local sensitivity is not smooth.

FORMULA / SENSITIVITY RULE

Piecewise rules can have jumps; derivative may not exist at the boundary.

IMPLEMENTATION NOTE

Log counts of threshold crossings during perturbation tests. Smooth position sizing can reduce cliffs if appropriate.

SYSTEM FIT

Threshold diagnostics protect against hidden fragility in backtests.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A discontinuous rule can look stable until data sits near the boundary.

Related: piecewise function, signal threshold, stop loss, position cap

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Marginal Contribution to Risk

INTERMEDIATE

ONE-SENTENCE MEANING

Marginal risk measures how portfolio risk changes when one weight changes slightly.

MEMORY JOGGER

What does one extra unit of this asset add to risk?

CORE IDEA

Portfolio risk is a function of weights, volatilities, and correlations. Marginal contribution identifies which positions increase or reduce total risk at the current allocation.

QUANT USE + MARKET EXAMPLE

Quant use: Use marginal risk for risk budgeting, position sizing, and portfolio diagnostics.

Example: A small allocation to a volatile asset may add less risk than expected if it diversifies existing positions.

FORMULA / SENSITIVITY RULE

For volatility $\sigma_p = \sqrt{w^T \Sigma w}$, marginal risk $i = \partial \sigma_p / \partial w_i$.

IMPLEMENTATION NOTE

Calculate covariance-based marginal risk and compare with historical scenario contributions.

SYSTEM FIT

Marginal risk belongs in portfolio construction and monitoring.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Marginal risk depends on the current portfolio, not just the standalone asset volatility.

Related: risk contribution, covariance, portfolio weights, diversification

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Marginal Contribution to Return

INTERMEDIATE

ONE-SENTENCE MEANING

Marginal return measures expected return change from a small allocation change.

MEMORY JOGGER

What extra return does the next dollar aim to buy?

CORE IDEA

Expected return sensitivity is easy to write but hard to estimate. The derivative may be simple, while the input forecast is uncertain.

QUANT USE + MARKET EXAMPLE

Quant use: Use marginal return with caution and always pair it with risk and estimation uncertainty.

Example: Increasing weight in a high-forecast asset raises expected portfolio return if the forecast is trusted.

FORMULA / SENSITIVITY RULE

Expected portfolio return $R = w^T \mu$. $\partial R / \partial w_i = \mu_i$ before constraints.

IMPLEMENTATION NOTE

Stress expected returns and compare allocation sensitivity under multiple forecast models.

SYSTEM FIT

Marginal return is an optimizer input, not a standalone decision rule.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Uncertain alpha estimates can dominate the allocation error.

Related: expected return, alpha, optimizer, estimation risk

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Portfolio Weight Sensitivity

INTERMEDIATE

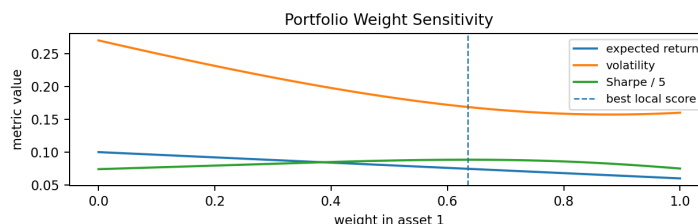
ONE-SENTENCE MEANING

Portfolio metrics change as weights change.

MEMORY JOGGER

Allocation is a function input.

PORTFOLIO CURVE



Changing weights changes return, volatility, and risk-adjusted score together.

CORE IDEA

Weights affect return, volatility, drawdown, turnover, and constraints. Sensitivity to weight changes helps reveal concentration risk and allocation fragility.

QUANT USE + MARKET EXAMPLE

Quant use: Use weight sensitivity to test whether a portfolio is robust to small rebalance or sizing changes.

Example: A portfolio whose risk jumps after a 2 percent weight shift may be sitting near a dangerous concentration boundary.

FORMULA / SENSITIVITY RULE

Metric sensitivity = partial metric / partial w_i , with constraints such as $\sum(w)=1$ handled explicitly.

IMPLEMENTATION NOTE

Bump each weight and rebalance the rest according to a defined rule. Otherwise the perturbation is ambiguous.

SYSTEM FIT + FAILURE MODE

System fit: Weight sensitivity belongs in portfolio construction, risk review, and deployment checks.

Confusion to avoid: Changing one weight requires specifying what happens to the other weights.

Related: constraints, marginal risk, risk parity, allocation

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Volatility Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Volatility sensitivity measures how model outputs respond to changes in volatility assumptions.

MEMORY JOGGER

What breaks when sigma changes?

CORE IDEA

Volatility affects position sizing, option values, risk estimates, stops, and regime filters. A model can look profitable under one volatility estimate and fragile under another.

QUANT USE + MARKET EXAMPLE

Quant use: Use volatility shocks to test risk models, strategies, and options books.

Example: A volatility-targeted strategy reduces exposure when estimated volatility rises.

FORMULA / SENSITIVITY RULE

Sensitivity to sigma: $\text{partial output} / \text{partial sigma}$. In options, this is closely related to vega.

IMPLEMENTATION NOTE

Shock realized volatility windows, EWMA decay, and implied volatility inputs separately.

SYSTEM FIT

Volatility sensitivity belongs in risk engines and regime-aware backtests.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Historical volatility and implied volatility are different inputs with different meanings.

Related: sigma, vega, risk targeting, regime

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Covariance and Correlation Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Covariance sensitivity measures response to changes in asset co-movement assumptions.

MEMORY JOGGER

Diversification depends on relationships, not just assets.

CORE IDEA

Portfolio risk can shift sharply when correlations rise during stress. A derivative with respect to covariance inputs asks how fragile the allocation is to relationship changes.

QUANT USE + MARKET EXAMPLE

Quant use: Use correlation shocks in portfolio stress testing and risk budgeting.

Example: Two assets that diversify in calm periods may become highly correlated during a crisis.

FORMULA / SENSITIVITY RULE

Portfolio variance = $w^T \Sigma w$. Sensitivity to Σ_{ij} is tied to $w_i w_j$ terms.

IMPLEMENTATION NOTE

Run correlation floors, crisis-correlation scenarios, and covariance shrinkage comparisons.

SYSTEM FIT

Covariance sensitivity belongs in portfolio validation.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Backtested diversification can vanish when correlations regime-shift.

Related: correlation, covariance matrix, diversification, stress test

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Beta Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Beta sensitivity measures exposure to a market or factor move.

MEMORY JOGGER

How much do you move with the benchmark?

CORE IDEA

Beta is a slope from regression: asset or strategy returns against factor returns. It is a derivative-like average sensitivity over a sample, not a permanent property.

QUANT USE + MARKET EXAMPLE

Quant use: Use beta to separate factor exposure from strategy-specific alpha.

Example: A supposedly market-neutral strategy with beta 0.35 may simply be long equity risk.

FORMULA / SENSITIVITY RULE

Regression slope $\beta = \text{Cov}(r_{\text{strategy}}, r_{\text{factor}}) / \text{Var}(r_{\text{factor}})$.

IMPLEMENTATION NOTE

Estimate rolling beta and stress it across regimes. Use robust regressions when outliers dominate.

SYSTEM FIT

Beta sensitivity belongs in performance attribution and risk control.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Low full-sample beta can hide high beta during crashes.

Related: factor model, alpha, regression slope, exposure

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Option Delta

INTERMEDIATE

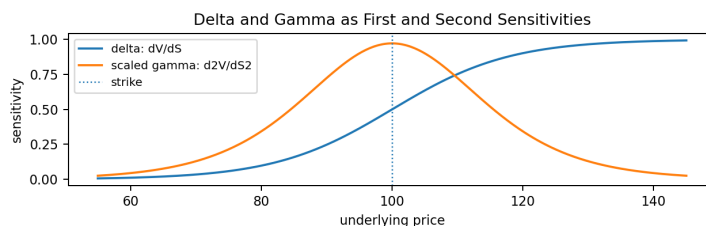
ONE-SENTENCE MEANING

Delta measures option value sensitivity to the underlying price.

MEMORY JOGGER

First-order option exposure.

GREEKS CHART



Delta is first-order option sensitivity; gamma shows how delta bends around the strike.

CORE IDEA

Delta is the derivative of option value with respect to underlying price. It changes across moneyness, time, and volatility.

QUANT USE + MARKET EXAMPLE

Quant use: Use delta to hedge or estimate small underlying-price effects on option value.

Example: A call option with delta 0.60 gains about 0.60 dollars for a small 1 dollar stock rise, before gamma effects.

FORMULA / SENSITIVITY RULE

Delta = $\partial V / \partial S$. Local estimate: $\Delta V \sim \Delta \cdot \Delta S$.

IMPLEMENTATION NOTE

Track delta by contract and aggregate portfolio delta by underlying. Recompute after large moves.

SYSTEM FIT + FAILURE MODE

System fit: Delta is a first-order risk exposure in options monitoring.

Confusion to avoid: Delta is not constant. It moves as the underlying, volatility, and time change.

Related: gamma, moneyness, hedge ratio, Greeks

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Option Gamma

INTERMEDIATE

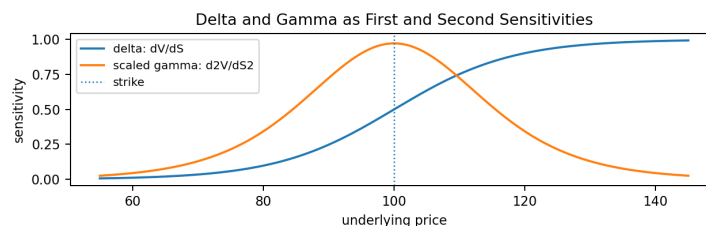
ONE-SENTENCE MEANING

Gamma measures how delta changes as the underlying price changes.

MEMORY JOGGER

Gamma is delta curvature.

GREEKS CHART



Gamma is curvature in option value with respect to underlying price.

CORE IDEA

Gamma is second-order sensitivity. It becomes especially important near strikes and near expiration. High gamma means local exposure can change fast.

QUANT USE + MARKET EXAMPLE

Quant use: Use gamma to understand convexity, hedge rebalancing needs, and nonlinear option risk.

Example: A delta hedge can become inaccurate quickly when gamma is high and the underlying moves.

FORMULA / SENSITIVITY RULE

Gamma = $\frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}$

IMPLEMENTATION NOTE

Monitor gamma concentration by strike and expiry. Stress large moves where delta-only estimates fail.

SYSTEM FIT + FAILURE MODE

System fit: Gamma belongs in nonlinear risk and hedging diagnostics.

Confusion to avoid: Gamma can help or hurt depending on position, premium, and realized movement.

Related: delta, convexity, second derivative, hedging

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Vega

INTERMEDIATE

ONE-SENTENCE MEANING

Vega measures option value sensitivity to volatility.

MEMORY JOGGER

What happens if implied volatility changes?

CORE IDEA

Vega is not a Greek letter in the classical alphabet naming sense, but in options it means volatility sensitivity. It can dominate PnL when implied volatility shifts.

QUANT USE + MARKET EXAMPLE

Quant use: Use vega for options books, volatility strategies, and event-risk trades.

Example: A long option position can lose even if price direction is right when implied volatility collapses.

FORMULA / SENSITIVITY RULE

Vega = $\partial V / \partial \sigma$. It is usually quoted per 1 volatility point.

IMPLEMENTATION NOTE

Aggregate vega across expiries and underlyings. Stress implied volatility up/down scenarios.

SYSTEM FIT

Vega belongs in options risk and volatility exposure monitoring.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Realized volatility and implied volatility are not the same; vega is about pricing volatility input.

Related: implied volatility, gamma, theta, volatility surface

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Theta Decay

INTERMEDIATE

ONE-SENTENCE MEANING

Theta measures option value sensitivity to the passage of time.

MEMORY JOGGER

Time passing changes option value.

CORE IDEA

Theta is often discussed as time decay, but sign and interpretation depend on position and model convention. Time sensitivity interacts with volatility and moneyness.

QUANT USE + MARKET EXAMPLE

Quant use: Use theta to understand carry, decay, and holding-period risk in options.

Example: A long out-of-the-money option can lose value each day even if the underlying does not move.

FORMULA / SENSITIVITY RULE

Theta = $\partial V / \partial t$, commonly reported as value change per day.

IMPLEMENTATION NOTE

Track theta exposure by expiry bucket. Compare expected decay with realized PnL.

SYSTEM FIT

Theta belongs in options carry analysis and trade lifecycle management.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Theta is not free income; short-theta positions often carry tail risk.

Related: time decay, option premium, gamma, vega

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Rho and Rate Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Rho measures option value sensitivity to interest-rate changes.

MEMORY JOGGER

Rates are an input too.

CORE IDEA

Rate sensitivity is often smaller than delta or vega for short-dated equity options, but it matters for longer-dated options, fixed income, and macro portfolios.

QUANT USE + MARKET EXAMPLE

Quant use: Use rate sensitivities for options, bonds, futures, swaps, and discounted cash-flow models.

Example: A long-dated option can change in value when discount rates shift.

FORMULA / SENSITIVITY RULE

$\text{Rho} = \partial V / \partial r$. Bond duration is another rate-sensitivity measure.

IMPLEMENTATION NOTE

Shock rate curves by parallel and non-parallel moves. Do not use one flat-rate bump for every instrument.

SYSTEM FIT

Rate sensitivity belongs in macro, fixed-income, and derivatives risk.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Different points on the yield curve can move differently; rho is often too simplified.

Related: duration, DV01, yield curve, discounting

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

DV01

INTERMEDIATE

ONE-SENTENCE MEANING

DV01 measures dollar value change for a one-basis-point yield move.

MEMORY JOGGER

Dollar pain per bp.

CORE IDEA

DV01 is a practical finite-difference sensitivity for fixed income. It converts rate movement into dollar risk.

QUANT USE + MARKET EXAMPLE

Quant use: Use DV01 to size and hedge bond or rate exposures.

Example: A portfolio with DV01 of 8,000 loses roughly 8,000 dollars for a 1 bp adverse parallel yield move, depending on sign convention.

FORMULA / SENSITIVITY RULE

$DV01 \sim [Price(y-1bp) - Price(y+1bp)]/2$, or quoted as dollar change per bp.

IMPLEMENTATION NOTE

Compute key-rate DV01 across curve tenors, not only parallel DV01.

SYSTEM FIT

DV01 belongs in fixed-income risk reporting and hedge construction.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Sign conventions vary; always state whether positive DV01 gains or loses when yields fall.

Related: duration, key-rate duration, yield curve, rho

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Bond Duration

INTERMEDIATE

ONE-SENTENCE MEANING

Duration measures bond price sensitivity to yield changes.

MEMORY JOGGER

Higher duration means more rate sensitivity.

CORE IDEA

Duration is a first-order approximation. It works for small yield moves but needs convexity for larger moves.

QUANT USE + MARKET EXAMPLE

Quant use: Use duration for bond risk, rate hedging, and macro exposure.

Example: A bond with duration 7 may lose roughly 7 percent for a 1 percent yield increase before convexity adjustment.

FORMULA / SENSITIVITY RULE

Approx percent price change $\sim -\text{Duration} * \Delta y$.

IMPLEMENTATION NOTE

Use modified duration for yield sensitivity and compute key-rate duration for curve-shape risk.

SYSTEM FIT

Duration is a first-order rate-risk measure.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Duration is not maturity, though longer maturity often increases duration.

Related: DV01, convexity, yield, fixed income

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Bond Convexity

INTERMEDIATE

ONE-SENTENCE MEANING

Convexity measures curvature of bond price with respect to yield.

MEMORY JOGGER

Duration changes when yields move.

CORE IDEA

Bond price-yield relationships are curved. Convexity improves the approximation for larger rate moves and explains why duration alone can misestimate risk.

QUANT USE + MARKET EXAMPLE

Quant use: Use convexity for rate stress tests and hedging nonlinear fixed-income exposure.

Example: Two bonds with similar duration can behave differently during large yield shocks because convexity differs.

FORMULA / SENSITIVITY RULE

Approx $\Delta P/P \sim -D \Delta y + 0.5C^*(\Delta y)^2$.

IMPLEMENTATION NOTE

Report duration and convexity together. Compare first-order and second-order shock estimates.

SYSTEM FIT

Convexity is second-order fixed-income sensitivity.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Positive convexity can be valuable, but price, carry, and optionality matter.

Related: duration, second derivative, DV01, rate risk

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Slippage Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Slippage sensitivity measures how strategy performance changes as execution price worsens.

MEMORY JOGGER

Edge minus execution friction.

CORE IDEA

Small slippage assumptions can dominate high-turnover strategies. Derivative thinking asks how much performance decays per unit of added slippage.

QUANT USE + MARKET EXAMPLE

Quant use: Use slippage sensitivity to reject strategies whose edge disappears under realistic fills.

Example: A strategy with 1.2 Sharpe at zero slippage and 0.1 Sharpe at 2 bps is highly execution-sensitive.

FORMULA / SENSITIVITY RULE

Cost slope = Delta performance / Delta slippage_bps.

IMPLEMENTATION NOTE

Run slippage ladders by asset, volatility regime, and trade size. Use conservative live assumptions.

SYSTEM FIT

Slippage sensitivity belongs in backtest validation and deployment readiness.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Zero-cost backtests are usually fantasy for active strategies.

Related: transaction costs, turnover, market impact, execution

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Transaction Cost Derivative

INTERMEDIATE

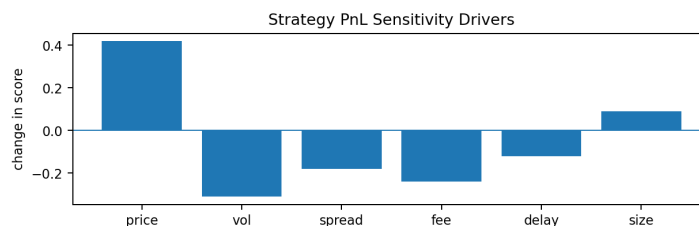
ONE-SENTENCE MEANING

Cost derivative measures how performance decays as fees, spread, or impact assumptions increase.

MEMORY JOGGER

Every bp of cost has a performance slope.

SENSITIVITY DRIVERS



Execution and cost assumptions can dominate net strategy quality.

CORE IDEA

Transaction costs are not just a line item; they are a sensitivity axis. High turnover can create a steep negative derivative with respect to cost assumptions.

QUANT USE + MARKET EXAMPLE

Quant use: Use cost derivatives to compare strategy robustness and capacity.

Example: A low-frequency strategy may lose little Sharpe per added bp, while a high-frequency strategy collapses.

FORMULA / SENSITIVITY RULE

$\text{partial metric} / \text{partial cost} < 0$ for most long-only cost increases. Estimate with finite differences.

IMPLEMENTATION NOTE

Compute net performance across cost grids and display break-even cost.

SYSTEM FIT + FAILURE MODE

System fit: Cost sensitivity is a mandatory validation report before trusting a backtest.

Confusion to avoid: Average cost may hide regime-specific spread widening and impact spikes.

Related: slippage, fee drag, turnover, capacity

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Capacity and Market Impact Sensitivity

INTERMEDIATE

ONE-SENTENCE MEANING

Capacity sensitivity measures how performance changes as trade size increases.

MEMORY JOGGER

More size can bend the cost curve.

CORE IDEA

Market impact is often nonlinear. Doubling size can more than double cost, especially in thin markets or stressed regimes.

QUANT USE + MARKET EXAMPLE

Quant use: Use size sensitivity to estimate realistic capital capacity.

Example: A microcap strategy may look excellent at tiny size but degrade sharply once order size becomes meaningful.

FORMULA / SENSITIVITY RULE

Impact often modeled as $\text{cost} = a * \text{size}^b$ with $b \geq 1$. Elasticity helps estimate scaling pain.

IMPLEMENTATION NOTE

Stress average daily volume participation, spread, volatility, and order slicing assumptions.

SYSTEM FIT

Capacity analysis belongs between backtest validation and live deployment.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Backtest return at small size does not imply scalable return.

Related: elasticity, market impact, liquidity, slippage

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Drawdown Sensitivity to Sequence

INTERMEDIATE

ONE-SENTENCE MEANING

Drawdown depends on the order of returns, not only their average.

MEMORY JOGGER

Same returns, different order, different pain.

CORE IDEA

A performance metric can be locally sensitive to where losses appear in the path. Derivative-like thinking helps test path perturbations and sequence fragility.

QUANT USE + MARKET EXAMPLE

Quant use: Use sequence tests to understand risk of clustering losses and bad timing.

Example: Moving several losses earlier in a return path can increase max drawdown even when total return is unchanged.

FORMULA / SENSITIVITY RULE

$\text{Drawdown}_t = \text{equity}_t / \text{running_max}_t - 1$. The running max creates path dependence and non-smooth behavior.

IMPLEMENTATION NOTE

Shuffle blocks, bootstrap paths, and stress loss clusters. Compare drawdown distribution, not only final CAGR.

SYSTEM FIT

Sequence sensitivity belongs in risk validation and strategy psychology checks.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Average return can look fine while path risk is unacceptable.

Related: path dependence, max drawdown, bootstrap, equity curve

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Risk Limits as Constraints

INTERMEDIATE

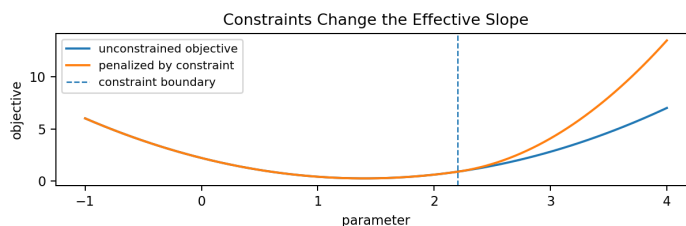
ONE-SENTENCE MEANING

Risk limits restrict the feasible set of model or portfolio choices.

MEMORY JOGGER

The optimizer can only move inside the fence.

CONSTRAINT CHART



Constraints can change the effective slope and block the unconstrained optimum.

CORE IDEA

Constraints change derivatives because movement may be blocked or redirected at boundaries. Leverage caps, sector caps, drawdown stops, and margin rules all affect local behavior.

QUANT USE + MARKET EXAMPLE

Quant use: Use constraint-aware sensitivity when evaluating portfolios or automated trading systems.

Example: If leverage is already at the cap, a positive signal cannot increase position size further.

FORMULA / SENSITIVITY RULE

Constrained optimization: minimize $J(\theta)$ subject to $g(\theta) \leq 0$. Active constraints alter the solution.

IMPLEMENTATION NOTE

Log which constraints are active at each rebalance or optimization step.

SYSTEM FIT + FAILURE MODE

System fit: Constraint status should be part of every allocation and execution report.

Confusion to avoid: Unconstrained gradients can recommend moves the system is not allowed to make.

Related: optimization, KKT conditions, leverage cap, feasible set

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Active vs Inactive Constraints

INTERMEDIATE

ONE-SENTENCE MEANING

An active constraint binds the solution; an inactive constraint does not currently affect it.

MEMORY JOGGER

A fence matters only when you are touching it.

CORE IDEA

Derivative behavior changes at active constraints. When a constraint binds, local moves may be clipped, redirected, or impossible.

QUANT USE + MARKET EXAMPLE

Quant use: Use active-constraint diagnostics to explain why an optimizer or sizing rule behaves unexpectedly.

Example: A portfolio may want more of an asset, but the sector cap is active, so weight cannot increase.

FORMULA / SENSITIVITY RULE

Active: $g(\theta)=0$ at the boundary. Inactive: $g(\theta)<0$ with room to move.

IMPLEMENTATION NOTE

Report active constraints and shadow prices when available. For rules-based systems, report clipped signals.

SYSTEM FIT

Active-constraint logs improve interpretability and risk governance.

CONFUSIONS + RELATED TERMS

Confusion to avoid: A flat allocation may reflect a binding constraint, not lack of alpha.

Related: feasible region, optimizer, position cap, shadow price

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Differentiability in Backtests

INTERMEDIATE

ONE-SENTENCE MEANING

Many backtest functions are not smooth enough for true derivatives.

MEMORY JOGGER

Backtests are often if-statements wearing math clothes.

CORE IDEA

Backtests include thresholds, fills, integer shares, stops, slippage rules, and conditional exits. The output can jump when a small input change alters trade timing.

QUANT USE + MARKET EXAMPLE

Quant use: Use finite perturbation and neighborhood analysis rather than assuming analytic derivatives exist.

Example: Changing a threshold by 0.01 may remove a trade that generated most of the historical profit.

FORMULA / SENSITIVITY RULE

Derivative may be undefined at discontinuities; use controlled finite differences and local grids.

IMPLEMENTATION NOTE

Count changed trades between baseline and perturbed runs. Large trade-set changes explain unstable sensitivity.

SYSTEM FIT

Backtest differentiability checks belong in validation.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Smooth-looking equity curves can come from non-smooth decision logic.

Related: threshold effect, finite difference, trade attribution, robustness

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Automatic Differentiation

INTERMEDIATE

ONE-SENTENCE MEANING

Automatic differentiation computes exact derivatives of differentiable code by applying chain rule internally.

MEMORY JOGGER

The computer tracks the chain rule for you.

CORE IDEA

Autodiff powers modern ML frameworks. It works for differentiable operations, but not for arbitrary backtest logic with discrete branches and external simulations unless designed carefully.

QUANT USE + MARKET EXAMPLE

Quant use: Use autodiff for neural nets, differentiable portfolio objectives, calibration, and some risk models.
Example: A volatility forecasting model can use autodiff to update weights by gradient descent.

FORMULA / SENSITIVITY RULE

Autodiff decomposes code into elementary operations and propagates derivatives through the computation graph.

IMPLEMENTATION NOTE

Use PyTorch, JAX, or TensorFlow for differentiable models; keep non-differentiable backtest steps outside or approximate them carefully.

SYSTEM FIT

Autodiff belongs in ML training and differentiable optimization components.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Autodiff is not the same as symbolic algebra or numerical finite difference.
Related: gradient, computation graph, ML optimization, chain rule

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Sensitivity Report Card

INTERMEDIATE

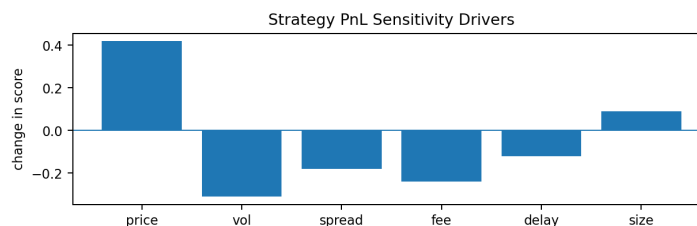
ONE-SENTENCE MEANING

A sensitivity report card summarizes which assumptions matter and how stable the results are.

MEMORY JOGGER

Do not trust one score; read the stress profile.

SENSITIVITY DRIVERS



A report card ranks which inputs move the strategy the most.

CORE IDEA

A professional research workflow reports baseline performance, local sensitivities, nonlinear stress results, parameter neighborhoods, cost decay, and path risk.

QUANT USE + MARKET EXAMPLE

Quant use: Use report cards to compare candidate strategies before deployment.

Example: Two strategies with the same Sharpe can differ completely in cost sensitivity and drawdown sensitivity.

FORMULA / SENSITIVITY RULE

Report: baseline metric, perturbation, Delta metric, slope, curvature clue, worst stress, stability rating.

IMPLEMENTATION NOTE

Generate standardized PDF/HTML/CSV sensitivity reports after every major backtest.

SYSTEM FIT + FAILURE MODE

System fit: This is the operational layer that turns calculus concepts into research governance.

Confusion to avoid: A report card is only useful if perturbations are realistic and reproducible.

Related: validation, robustness, diagnostics, model governance

EXTRA DIAGNOSTIC

A usable derivative page should leave you knowing what variable was perturbed, by how much, what changed, and whether the result is stable under a...

Robustness Workflow

INTERMEDIATE

ONE-SENTENCE MEANING

A robustness workflow tests whether model behavior survives nearby changes and realistic stress.

MEMORY JOGGER

Make the edge prove it is not a needle.

CORE IDEA

Derivative, sensitivity, and curvature tools support one goal: deciding whether a model is stable enough to trust. Robustness is not one test; it is a sequence of local, grid, scenario, cost, and path checks.

QUANT USE + MARKET EXAMPLE

Quant use: Use robustness workflow before allocating capital, scaling size, or automating a strategy.

Example: A strategy passes only if it survives nearby parameters, higher costs, delayed fills, volatility regimes, and out-of-sample data.

FORMULA / SENSITIVITY RULE

Robust decision = good baseline + stable neighborhood + controlled downside under stress + explainable sensitivity drivers.

IMPLEMENTATION NOTE

Pipeline: baseline -> local bumps -> parameter grid -> cost ladder -> path bootstrap -> regime split -> deployment checklist.

SYSTEM FIT

This page is the final system-level integration of derivative thinking.

CONFUSIONS + RELATED TERMS

Confusion to avoid: Robustness does not mean guaranteed future performance; it means fewer obvious fragility failures.

Related: sensitivity, validation, out-of-sample, deployment

QUICK RECALL

State the input, output, units, local step size, and whether the derivative is analytic, numeric, or only an approximation.

Formula Summary Sheet

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HOW TO READ FORMULAS

Each formula is a diagnostic template. Before using it, identify the input, output, perturbation unit, smoothness assumption, and whether the estimate is analytic or numerical.

DERIVATIVE FORMULA MAP

Formula	Expression	Use
Derivative	$f_{\text{prime}}(x)=\lim_{(h \rightarrow 0)}[f(x+h)-f(x)]/h$	local slope
Finite diff	$[f(x+h)-f(x)]/h$	practical estimate
Central diff	$[f(x+h)-f(x-h)]/(2h)$	diagnostic accuracy
Second derivative	$d/dx[f_{\text{prime}}(x)]$	curvature
Gradient	$[\text{partial } f/\text{partial } \theta_{\text{i}}]$	multi-parameter slope
Directional	$\text{grad } f \text{ dot } u$	scenario path
Taylor 1	$f(x+h)-f(x)+f_{\text{prime}}(x)h$	linear approx
Taylor 2	add $0.5 f_{\text{double_prime}}(x)h^2$	curvature approx
Elasticity	$(dy/dx)(x/y)$	percent response
DV01	dollar change per bp	rate risk

PRACTICAL RULE

For research systems, exact symbolic elegance matters less than reproducible perturbation tests and honest interpretation of approximation error.

RECALL QUESTION

Which formula is only valid locally, which one needs a finite step size, and which one becomes dangerous when the denominator is near zero?

HOW TO USE THIS PAGE

Turn each item into a test: perturb one input, record output change, compare neighboring perturbations, and decide whether the result is robust enough to inform a model decision.

Failure-Mode Table

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PURPOSE

Most derivative mistakes in quant research are not calculus mistakes. They are data, units, leakage, discontinuity, cost, or validation mistakes.

COMMON SENSITIVITY FAILURES

Failure	Symptom	Fix
Look-ahead difference	central feature uses future data	use backward/causal features
Too-small h	slope jumps wildly	sweep h values
Threshold cliff	tiny input flips trade	count changed trades
Cost fragility	edge dies with fees	run cost ladder
Isolated peak	one hot parameter cell	use neighborhood score
Noisy curvature	second derivative spikes	smooth/test robustness
Ratio illusion	denominator near zero	show numerator/denominator
Regime hiding	full-sample slope stable	split by regime

BEST DIAGNOSTIC

When a sensitivity result looks impressive, rerun it nearby. Change h , sample window, cost level, regime split, and seed. Fragile results reveal themselves through inconsistency.

DEPLOYMENT GATE

No live model should rely on a single local derivative, single parameter optimum, or single optimistic cost assumption.

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Diagnostic Interpretation Table

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SYMPTOMS AND INTERPRETATIONS

Observation	Interpretation	Next step
High score, steep local slope	possibly overfit parameter	check neighborhood median
Low score sensitivity to costs	execution-robust	still test liquidity regimes
Large gamma / convexity	nonlinear exposure	use stress not delta-only
Flat parameter surface	robust or inactive	check trade count and signal use
Sensitivity sign flips	unstable relationship	split samples/regimes
Metric improves as vol rises	vol-linked behavior	inspect exposure and sizing
Drawdown shifts under shuffle	sequence fragile	bootstrap and cap risk
High beta during stress	hidden market risk	rolling factor analysis

INTERPRETATION RULE

A diagnostic is not a verdict. It is a pointer to the next test. Sensitivity analysis should narrow uncertainty, not create a false sense of certainty.

SYSTEM-DESIGN CONNECTION

Store diagnostic outputs in a comparable schema so future models can be ranked by robustness, not just by headline return or Sharpe.

QUICK RECALL

If the local result changes under a neighboring perturbation, the model is telling you the chosen point is not enough information.

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Practice Lab: Build the Sensitivity Toolkit

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GOAL

Convert this manual into reusable research functions. Each exercise should output a small table and a chart that can be attached to a model review.

IMPLEMENTATION EXERCISES

Exercise	Action	Output
Cost ladder	Run 0, 2, 5, 10 bps	find break-even cost
Parameter bump	Shift lookback +/- 10%	measure neighborhood stability
Trade-set comparison	Count changed trades	detect threshold cliffs
Delta-gamma estimate	Compare first/second order	see curvature error
Portfolio bump	Move 1% weight	find marginal risk
Regime split	Bull, bear, high vol	check slope consistency
Path shuffle	Block bootstrap returns	inspect drawdown sensitivity
Scenario vector	Shock correlated inputs	compare to partials

MINIMUM REPORT SCHEMA

model_id, baseline_config_hash, perturbation_name, perturbation_size, output_metric, baseline_value, perturbed_value, sensitivity, sample_window, notes.

QUALITY CHECK

A good tool makes it easier to reject fragile strategies. It should not be designed to make every backtest look defensible.

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Index and Concept Map

INTERMEDIATE

HOW THE MANUAL CLUSTERS

The concepts form a path: local change -> numerical estimation -> curvature -> multivariable sensitivity -> approximation -> optimization -> market risk -> robustness reporting.

CONCEPT CLUSTERS

Cluster	Key terms	Where
Derivative mechanics	derivative, secant, tangent, delta notation, units, h selection	Pages 8-17
Discrete estimates	first differences, finite differences, noise, smoothing	Pages 14-19
Curvature	second derivative, acceleration, convexity, concavity	Pages 20-23
Multivariable tools	partials, gradient, directional derivative, Jacobian, Hessian	Pages 24-28
Approximation	chain rule, product rule, quotient rule, elasticity, Taylor	Pages 29-36
Optimization	numeric tests, gradients, learning rates, flat/exploding gradients, thresholds	Pages 37-47
Risk and portfolio	marginal risk/return, weights, volatility, covariance, beta	Pages 48-53
Derivatives markets	delta, gamma, vega, theta, rho, DV01, duration, convexity	Pages 54-61
Backtest robustness	slippage, costs, capacity, drawdown, constraints, autodiff, report cards	Pages 62-71

FINAL MEMORY HOOK

Derivative thinking means never trusting a single number without asking how it changes nearby. In quant research, the neighborhood often matters more than the point.

NEXT VOLUMES TO PAIR

Optimization, Probability Distributions, Regression, Time Series, Portfolio Risk, Options Pricing, Backtesting, and Execution Cost Modeling.

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